

Evidence 1: Highly Available & Globally Scalable Web Application (AWS)

Candidate: Jehanzeb Bashir

Role: AWS Solutions Architect (Professional Level)

Organization: Athinaa Ltd (United Kingdom)

Project Summary

As part of my role at **Athinaa Ltd (United Kingdom)**, I led the design and implementation of a highly available and globally scalable cloud architecture on **AWS**.

This project was delivered for a **production workload** through a client engagement with **Cisco Ltd**, where I was responsible for modernising a legacy system that was experiencing downtime and performance limitations.

The solution was designed to support high concurrent user traffic, eliminate single points of failure, and ensure uninterrupted service availability across regions.

My Role & Leadership

I acted as the **Lead Architect and Primary Engineer**, responsible for the end-to-end design, implementation, and optimization of the system.

- Architected a **multi-region, multi-AZ cloud architecture** for high availability
- Made key design decisions for **load balancing, failover strategies, and database reliability**
- Designed secure network isolation using **VPC segmentation and subnet design**
- Implemented monitoring, alerting, and operational best practices
- Led performance optimization and cost-efficiency initiatives

I independently made **critical architectural decisions** that directly improved system availability, scalability, and security, demonstrating strong technical leadership.

Architecture & Technical Design

The platform follows a **multi-tier, globally distributed cloud architecture**:

- **Global Layer:** Amazon CloudFront and Route 53 for low-latency routing and DNS-based failover
- **Application Layer:** Application Load Balancer distributing traffic across Auto Scaling EC2 instances
- **Data Layer:** Amazon RDS (Multi-AZ) with automated failover for high availability
- **Caching Layer:** Amazon ElastiCache (Redis) to reduce database load and improve response time
- **Security Layer:** AWS WAF, IAM, and security groups enforcing least-privilege access

This architecture ensures **high performance, fault tolerance, and seamless scalability across regions**.

Technical Complexity

- Designed a **globally distributed architecture** with automated failover capabilities
- Implemented **Auto Scaling** based on real-time traffic patterns
- Engineered secure **VPC architecture with subnet isolation**
- Optimized performance using **CDN caching (CloudFront) and in-memory data stores (Redis)**

- Configured **Layer 7 load balancing (ALB)** for intelligent traffic routing
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Challenges & Architectural Decisions

- Designed a **multi-region failover strategy**, ensuring minimal downtime during regional outages
- Optimized Route 53 health checks and failover policies to achieve **failover time under 5 minutes (RTO)**
- Solved **database replication and consistency challenges**, achieving near-zero data loss (RPO)
- Tuned Auto Scaling policies to efficiently handle unpredictable traffic spikes
- Balanced **performance vs cost trade-offs** using CloudFront caching and ElastiCache

These decisions ensured a **highly resilient, scalable, and cost-optimized system** capable of handling real-world production workloads.

High Availability & Disaster Recovery

- Deployed infrastructure across **multiple availability zones and regions**
 - Implemented **Route 53 health checks** for automated failover
 - Achieved **Recovery Time Objective (RTO) < 5 minutes**
 - Achieved **Recovery Point Objective (RPO) near zero**
 - Eliminated single points of failure across all application layers
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Impact & Measurable Results

- Achieved **99.95% system availability**, ensuring uninterrupted user access
 - Reduced application response time by **60%**, significantly improving user experience
 - Enabled the system to handle **5× traffic growth** without downtime or performance degradation
 - Reduced infrastructure costs by **30–40%** through Auto Scaling and optimization
 - Improved global performance through low-latency content delivery via CDN
 - Ensured **business continuity** with robust multi-region disaster recovery capabilities
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Commercial Context & Verification

This project was executed under a professional services engagement between Athinaa Ltd and Cisco Ltd.

The work can be validated through:

- Cisco Ltd confirmation letter
 - API Gateway and Lambda configuration evidence
 - CloudWatch logs and system monitoring dashboards
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Conclusion

This project demonstrates my ability to design and deliver **enterprise-grade, globally scalable cloud architectures** with a strong focus on availability, performance, and security.

It highlights my expertise in:

- Distributed system design
- High availability and disaster recovery
- Performance optimization at scale

The measurable improvements in system reliability, scalability, and user experience demonstrate my ability to deliver **high-impact cloud solutions that drive real business value**, positioning

DIAGRAM PREPARED BY JEHAZEB BASHIR AWS SOLUTION ARCHITECT CERTIFIED PROFESSIONAL

